ZIHOU REN

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EDUCATION

Tsinghua University — *Ph.D. Student, Cyberspace Security* *Sep. 2024–Jul. 2029*
Institute for Network Sciences and Cyberspace; GPA: 4.0/4.0.
Research interests: systems security, container security, and kernel performance profiling.

Tsinghua University — *B.Eng., Computer Science and Technology* *Sep. 2020–Jul. 2024*
Department of Computer Science and Technology; GPA: 3.8/4.0; top 30% of the department.
Tsinghua University Scholarship for Excellence in Science and Technology Innovation.

INTERNSHIPS

Shanghai SiOrigin Technology Co., Ltd. *Sep. 2025–May 2026*
RISC-V compiler middle-end and back-end development and performance optimization:

- Implemented the full translation pipeline from builtins to assembly for 1,000+ custom instructions.
- Designed hardware-aware dataflow analysis and peephole optimizations to improve cache utilization and computational performance.
- Profiled and optimized compiler bottlenecks, reducing compilation time by nearly 20%.

Shanghai Zhizhenjunzhi Technology Co., Ltd. *Jun. 2023–Jul. 2023*
Researched unknown network protocol identification; built a feature extraction and classification system using clustering and deep learning (CNNs, LSTMs) for noisy, high-dimensional network traffic sequences.

RESEARCH

Network and System Reliability and Security Lab, Tsinghua University *Apr. 2023–Present*
Advisor: Associate Professor Long Wang

- **Z. Ren**, *et al.* No Touching Others' Assets: Detecting Container Escapes via CAGEGuard. *Submitted to USENIX Security (**CCF-A**).* Modeled container boundaries through asset ownership and detected escapes using eBPF; achieved a 90% detection rate with low false positives and overhead.
- **Z. Ren**, *et al.* RootSeeker: Evidence-Guided Root Cause Analysis for Container Escapes. *Submitted to Journal of Software (**CCF Chinese-A**).* Developed self-correction mechanisms and agent-oriented log and code search engines to analyze escape logs and locate source-code vulnerabilities; achieved 48% end-to-end function-level accuracy.
- C. Liu, **Z. Ren**, *et al.* Understanding eBPF Performance Overhead. *Submitted to EuroSys (**CCF-A**).* Systematically measured eBPF performance overhead.
- Ongoing: security isolation for AI agents, defining task-specific boundaries for the assets an agent may access.

Intelligent Technology and Systems Lab, Tsinghua University *Sep. 2022–Mar. 2023*
Advisor: Professor Yiqun Liu

- Studied user click models and reinforcement learning for ranking tasks.

Institute of HCI and Media Integration, Tsinghua University *Sep. 2021–Jun. 2022*
Advisor: Professor Jiangtao Wen

- X. Zhang, **Z. Ren**, *et al.* Phocus: Picking Valuable Research from a Sea of Citations. Used neural networks to assess citation importance and develop a new research evaluation framework (arXiv:2201.02915).

SKILLS

Programming: Python, C/C++, Rust, JavaScript, HTML/CSS
Vibe Coding: Codex, Claude Code, skill authoring
Machine Learning: PyTorch **High-Performance Computing:** eBPF, DPDK, perf
Tools: Git, CMake, LLVM, Docker, LaTeX **English:** CET-6, 649/710

LEADERSHIP AND SERVICE

Tsinghua University Ski Team — *Captain* *Mar. 2025–Feb. 2026*
Undergraduate Compilers Course — *Teaching Assistant* *Sep. 2024–Present*
Tsinghua University Student Ski Association — *President* *Sep. 2023–Aug. 2024*